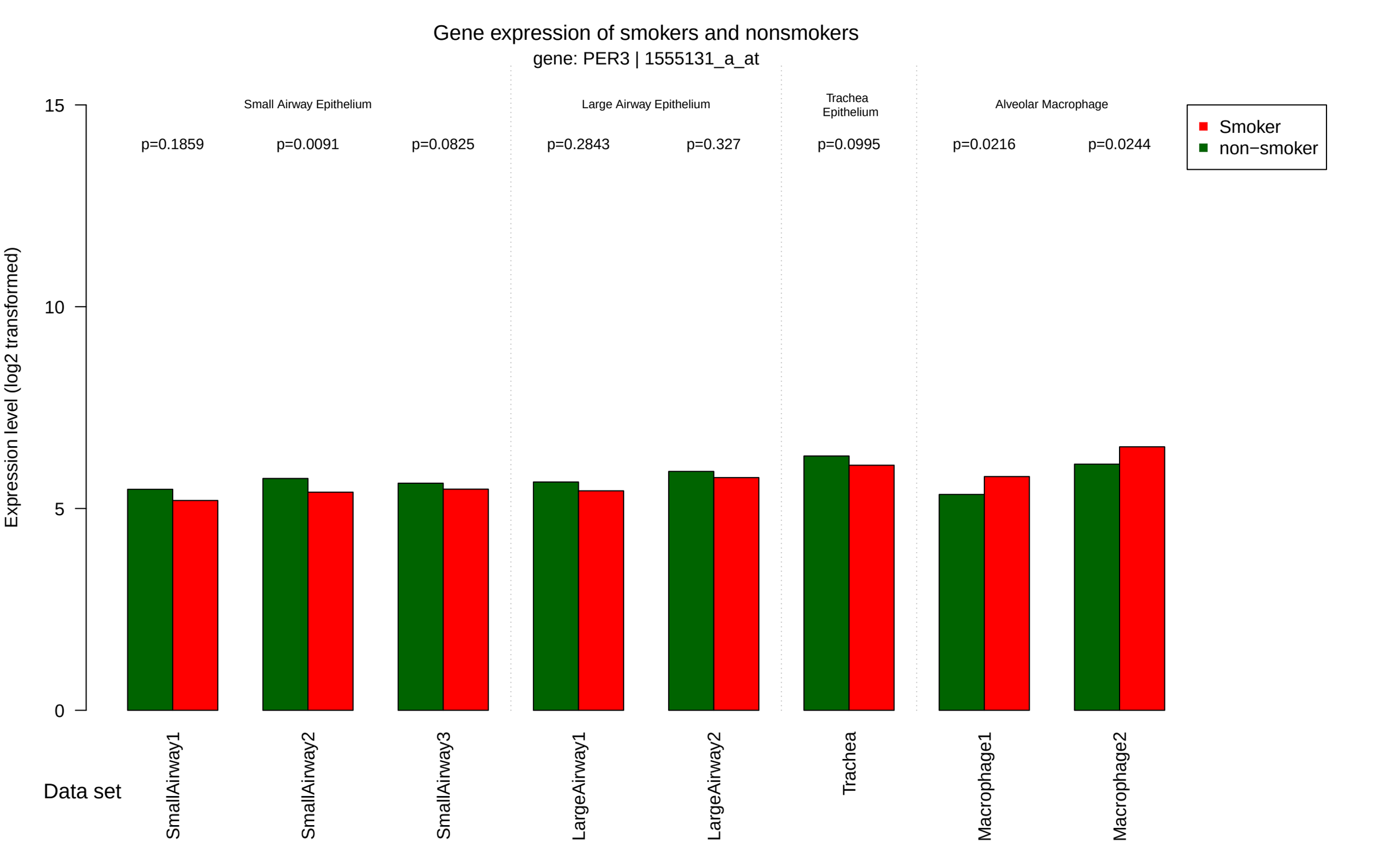
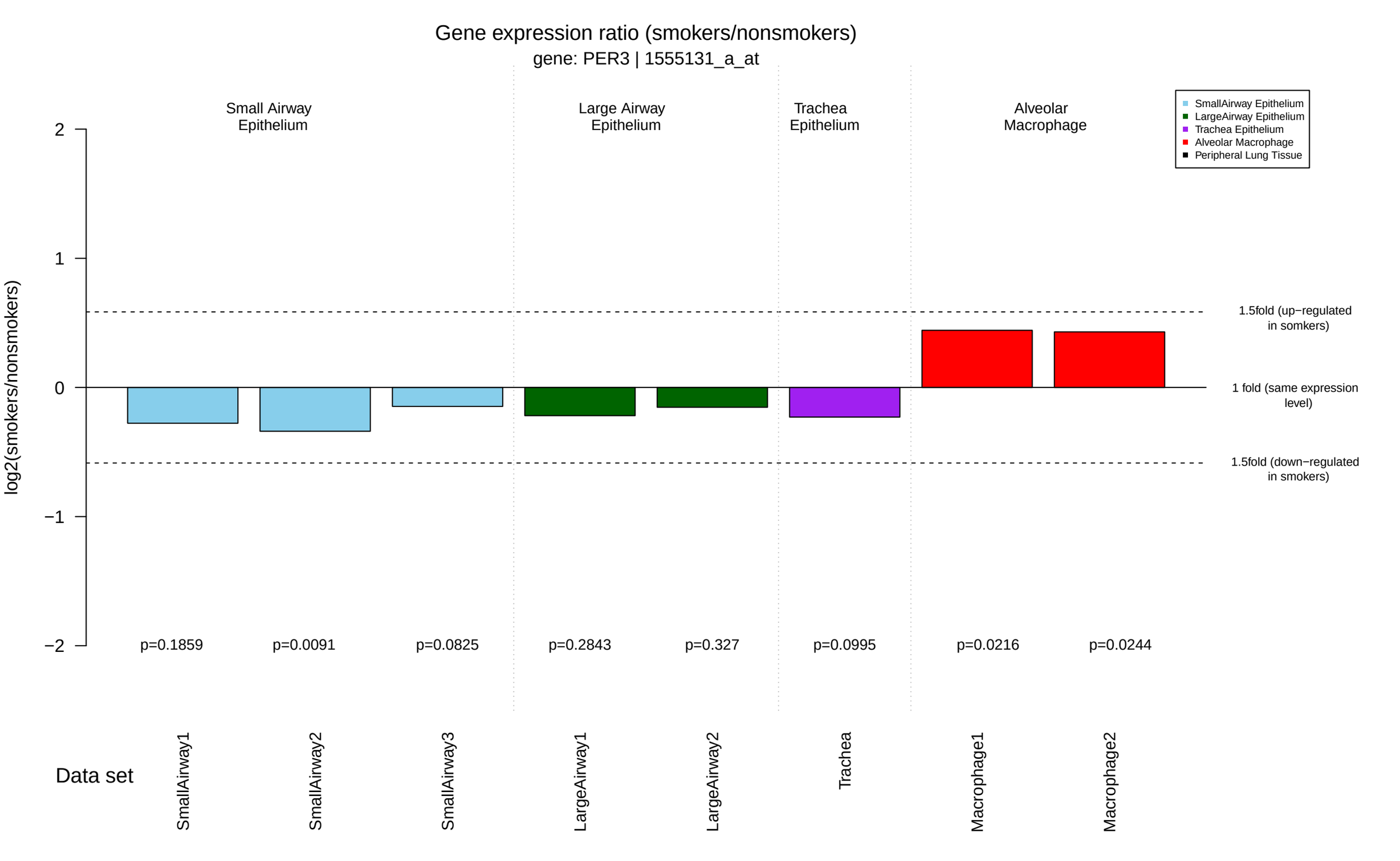
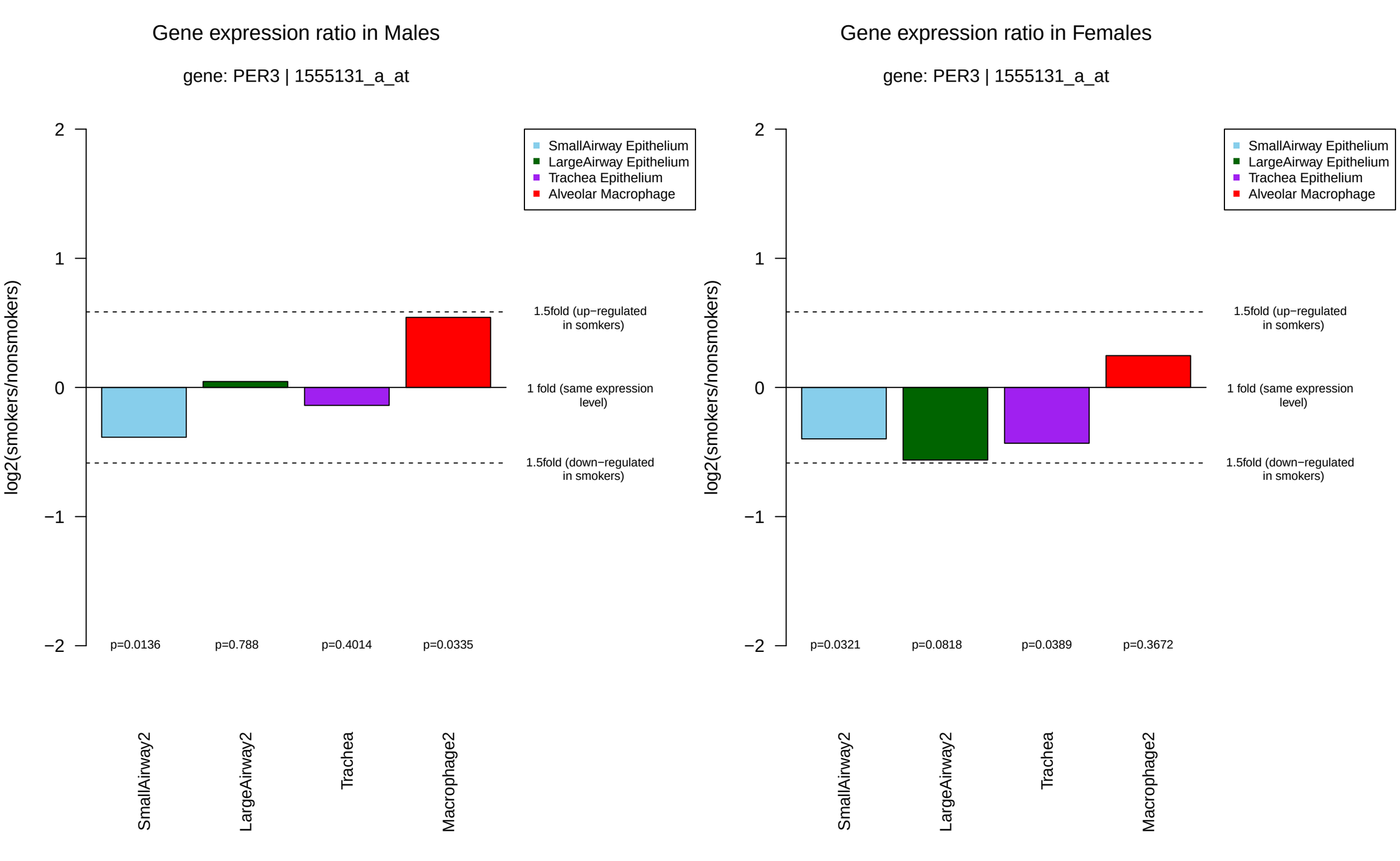
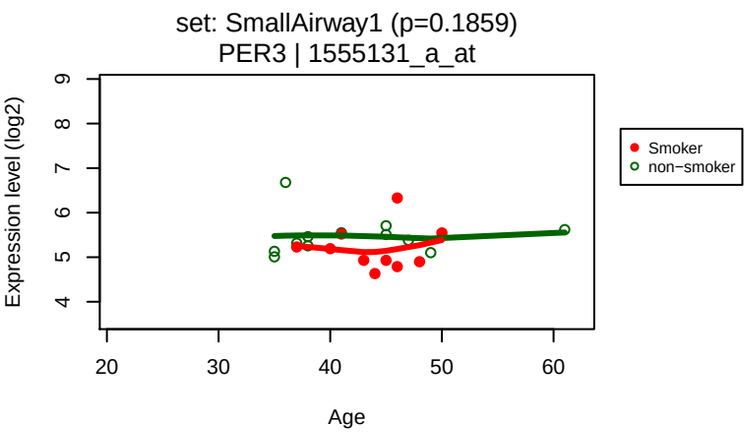
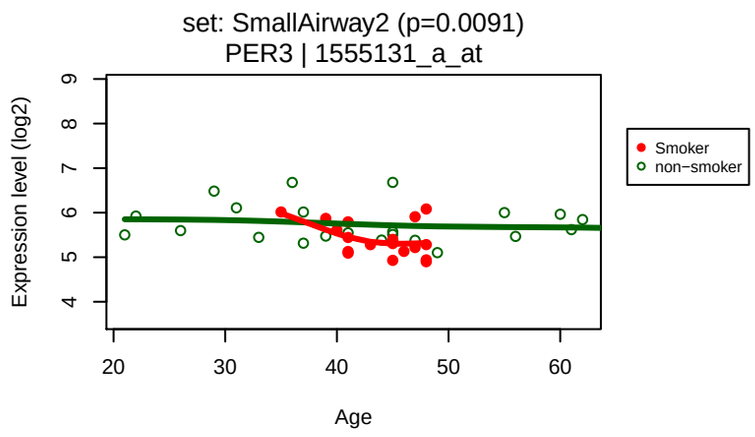
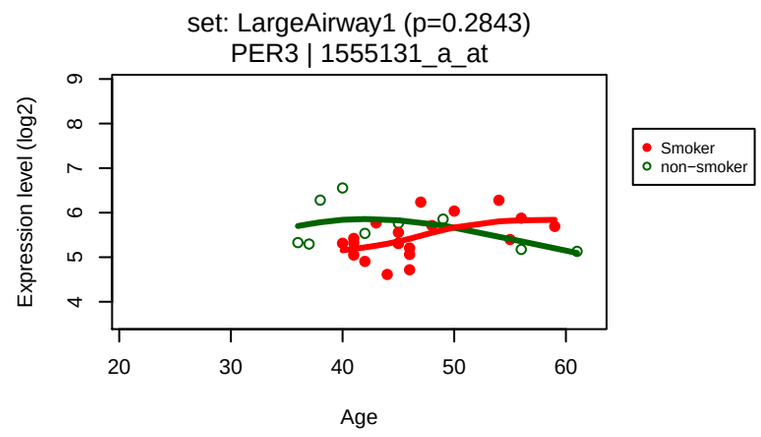

Gene expression of smokers and nonsmokers
gene: PER3 | 1555131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1859
p=0.0091
p=0.0825
p=0.2843
p=0.327
p=0.0995
p=0.0216
p=0.0244
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

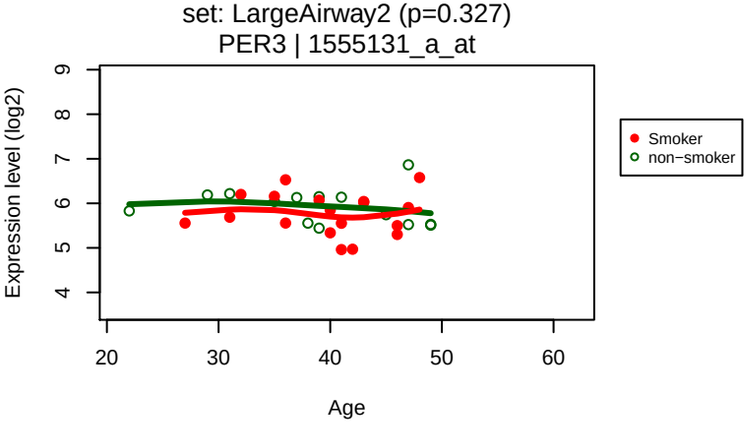

Gene expression ratio (smokers/nonsmokers)
gene: PER3 | 1555131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1859
p=0.0091
p=0.0825
p=0.2843
p=0.327
p=0.0995
p=0.0216
p=0.0244
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

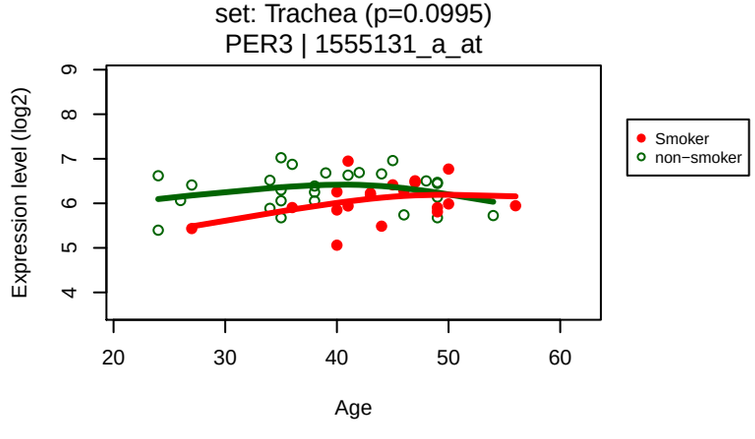

Gene expression ratio in Males
gene: PER3 | 1555131_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0136
p=0.788
p=0.4014
p=0.0335
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER3 | 1555131_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0321
p=0.0818
p=0.0389
p=0.3672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

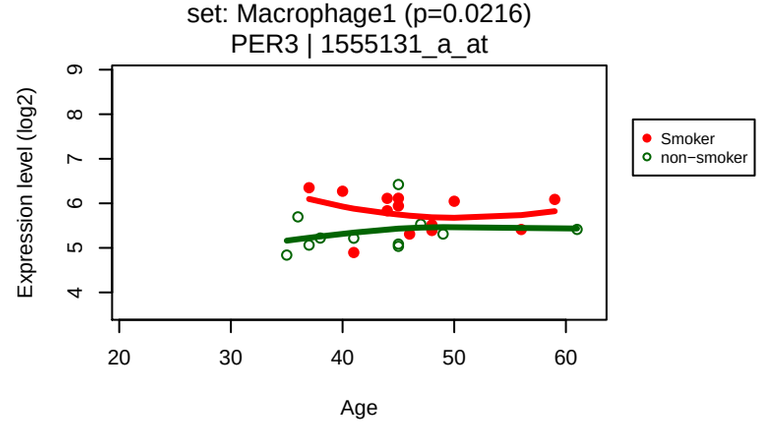

set: SmallAirway1 (p=0.1859)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker

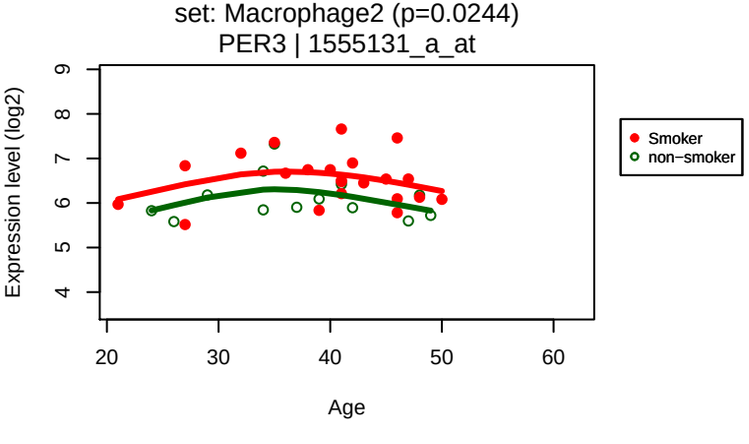

set: SmallAirway2 (p=0.0091)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker

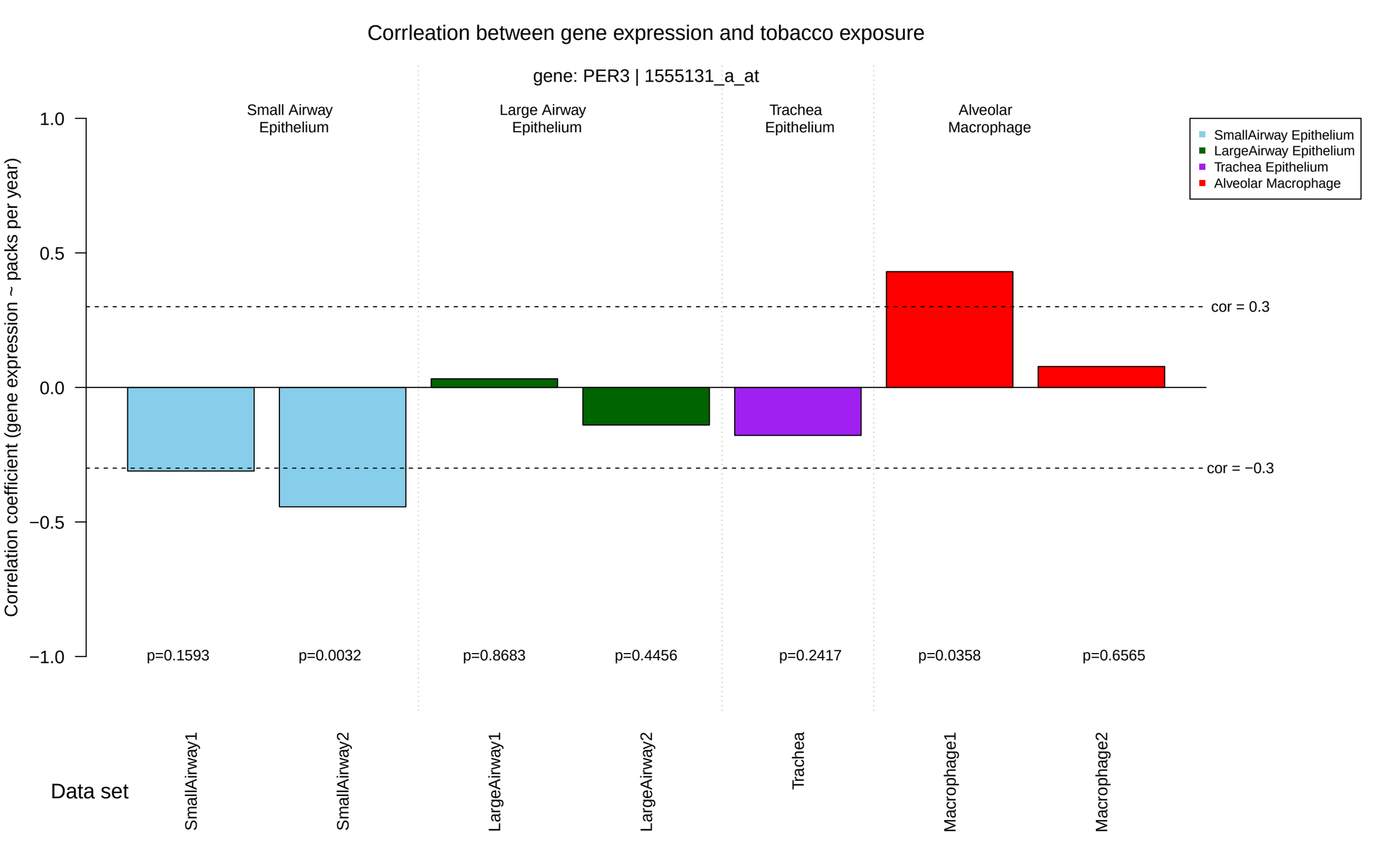

set: LargeAirway1 (p=0.2843)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.327)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0995)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0216)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0244)
PER3 | 1555131_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER3 | 1555131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1593
p=0.0032
p=0.8683
p=0.4456
p=0.2417
p=0.0358
p=0.6565
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2